

partitions-leanblueprint

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0.1 Definitions

Definition 1 (Integer Modular Form). An integer modular form of weight $k \in \mathbb{N}$ is a sequence $a : \mathbb{N} \rightarrow \mathbb{Z}$ such that $\sum_{n=0}^{\infty} a(n)q^n$ is a modular form of weight k , where $q = e^{2\pi iz}$.

Theorem 2. $\frac{2k}{\text{bernoulli}_{2k}}$ is an integer

Definition 3 (Eisenstein Series). For $k \geq 2 \in \mathbb{N}$, the Eisenstein series E_k is an integer modular form of weight $2k$ defined by

$$E_k = 1 - \frac{2k}{B_{2k}} \sum_{n=1}^{\infty} \sigma_{k-1}(n)q^n$$

where B_k is the k th Bernoulli number and $\sigma_k(n)$ is the sum of the k th powers of the divisors of n .

Theorem 4.

For $n > 0$, ℓ divides the n th coefficient in the q -expansion of $E_{\ell-1}$.

Definition 5 (Modular Form Mod ℓ). A modular form mod ℓ of weight $k \in \mathbb{Z}/(\ell-1)\mathbb{Z}$ is a sequence $a : \mathbb{N} \rightarrow \mathbb{Z}/\ell\mathbb{Z}$ such that there exists an integer modular form b of weight k' where $b \equiv a \pmod{\ell}$ and $k' \equiv k \pmod{\ell-1}$.

Definition 6 (Theta). Θ sends modular forms mod ℓ of weight k to weight $k+2$ by $(\Theta a)n = na(n)$.

Definition 7 (U Operator).

The operator U sends modular forms mod ℓ of weight k to weight k by $(a|U)n = a(\ell n)$.

Definition 8 (hasWeight). A modular form mod ℓ called a has weight $j \in \mathbb{N}$ if there exists an integer modular form b of weight j such that $b \equiv a \pmod{\ell}$.

Definition 9 (Filtration). Let a be a modular form mod ℓ . The filtration of a , $\omega(a)$, is defined as the minimum natural number j such that a has weight j . The filtration of the zero function is 0.

Definition 10 (multiplication and exponentiation). It's worth stating how multiplication and exponentiation are defined here, because they are not defined in the normal way. The multiplication of two modular forms mod ℓ called a and b is defined as

$$(a \cdot b)n = \sum_{x+y=n} a(x)b(y).$$

The exponentiation of a modular form mod ℓ called a to the power of $j \in \mathbb{N}$ is defined as

$$(a^j)n = \sum_{x_1+\dots+x_j=n} \prod_{i=1}^j a(x_i).$$

Theorem 11 (Pow Prime). Let ℓ be a prime and a a modular form mod ℓ of any weight. Then

$$a^\ell(n) = \begin{cases} a(n/\ell) & \text{if } \ell \mid n \\ 0 & \text{otherwise} \end{cases}$$

Proof.

□

0.2 Theorems

Theorem 12. *Let a be a modular form mod ℓ . Then $(a|U)^\ell = a - \Theta^{\ell-1}a$.*

Proof.

□

Theorem 13.

If an integer modular form of a weight k reduces to a modular form mod ℓ of weight j then we must have $k \equiv j \pmod{\ell-1}$

Theorem 14. *Let a be a modular form mod ℓ of weight k . Then $\omega(a) \equiv k \pmod{\ell-1}$.*

Proof.

□

Definition 15 (Delta).

Δ is the sequence obtained from $q(\prod_{n=1}^{\infty} (1 - q^n))^{24}$. It is an integer modular form of weight 12. The modular form mod ℓ , also called Δ , is defined to be the reduction of Δ .

Definition 16. For a prime $\ell \geq 5$, we define $\delta_\ell = \frac{\ell^2-1}{24} \in \mathbb{N}$. We define $f_\ell = \Delta^{\delta_\ell}$, which is an integer modular form of weight $12\delta_\ell$. We also define the reduction of f_ℓ as an modular form mod ℓ

Definition 17. For $k \in \mathbb{N}$, define $\dim(k) = \lfloor k/6 \rfloor + \begin{cases} 0 & \text{if } k \equiv 1 \pmod{6} \\ 1 & \text{else} \end{cases}$.

$\dim(k)$ is the dimension of the space of modular forms of weight $2k$.

Definition 18.

For $k \neq 1$ and $c < \dim(k)$, write k uniquely as $k = 6\ell + k'$, where $k' \in \{0, 2, 3, 4, 5, 7\}$. $G_{k,c}$ is a modular form of weight $2k$ defined as

$$G_{k,c} = E_2^a E_3^b E_2^{\ell-3c} \Delta^c$$

where a, b are the unique natural numbers with $2a + 3b = k'$.

We have that the set of all such $G_{k,c}$ for fixed k form a basis for the modular forms of weight $2k$.

Definition 19 (ord). Let b be an integer modular form. The order of b , $\text{ord}(b)$, is defined as the minimum $n \in \mathbb{N}$ such that $b(n) \neq 0$. This is the order of vanishing of b at infinity.

Theorem 20. $\text{ord}(a \cdot b) = \text{ord}(a) + \text{ord}(b)$.

Proof.

□

Theorem 21. $(a \cdot b)(\text{ord}(a) + \text{ord}(b)) = a(\text{ord}(a)) \cdot b(\text{ord}(b))$.

Proof.

□

Theorem 22. $\text{ord}(\Delta) = 1$.

Proof.

□

Theorem 23. For any k , $\text{ord}(E_k) = 0$.

Proof.

□

Theorem 24. $\text{ord}(G_{k,c}) = c$.

Proof.

□

Theorem 25. $G_{k,c}(c) = 1$.

Proof. □

Theorem 26. *If the coefficients of a and b match up to $\dim(k)$, then $a = b$.*

Proof. □

Theorem 27. *Let a be a modular form of weight $2k$. If $\text{ord}(a) = \dim(k) - 1$ then $a = a(\dim(k) - 1)G_{k, \dim(k)-1}$.*

Proof. □

Theorem 28. *Let a be a modular form mod ℓ of weight k . If $\text{ord}(a) \geq m$, then there exists an integer modular form b of weight k such that*

- (1) $\text{ord}(b) \geq m$, and
- (2) the reduction of b is equal to a .

Proof. □

Lemma 29. $\omega(f_\ell) = 12\delta_\ell = \frac{\ell^2-1}{2}$.

Proof. □

Theorem 30. *This is part (1) of Lemma 2.1.*

Let a be a modular form mod ℓ . Then $\omega(\Theta a) \leq \omega(a) + \ell + 1$.

Theorem 31. *This is part (2) of Lemma 2.1.*

Let a be a modular form mod ℓ . Then $\omega(\Theta a) = \omega(a) + \ell + 1$ if and only if $\ell \nmid \omega(a)$.

Theorem 32. *This is Lemma 3.2.*

For all $m \in \mathbb{N}$, $\omega(\Theta^m f_\ell) \geq \omega(f_\ell) = \frac{\ell^2-1}{2}$.

Proof. □

Theorem 33. *The conclusion of Lemma 3.3*

If $f_\ell|U = 0$ then $\omega(\Theta^{\ell-1} f_\ell) = \frac{\ell^2-1}{2}$

Proof. □

Lemma 34. *Let a be a modular form mod ℓ . If $\ell \mid \omega(a)$ then there exists an $\alpha \in \mathbb{N}$ such that $\omega(\Theta a) = \omega(a) + \ell + 1 - (\alpha + 1)(\ell - 1)$.*

Proof. □

Theorem 35. *If $(f_\ell|U) = 0$ then $\ell \mid \omega(\Theta^{\ell-2} f_\ell)$.*

Proof. □

Lemma 36. *For all $m \in \mathbb{N}$ with $m \leq \frac{\ell+1}{2}$, $\omega(\Theta^m f_\ell) = \frac{\ell^2-1}{2} + m(\ell + 1)$.*

Proof. Induction. □

Theorem 37. $\ell \mid \omega(\Theta^{\frac{\ell+1}{2}} f_\ell)$.

Proof. □

Definition 38. We define α to be the natural number such that $\omega(\Theta^{\frac{\ell+3}{2}} f_\ell) = \frac{\ell^2-1}{2} + \frac{\ell+3}{2}(\ell + 1) - (\alpha + 1)(\ell - 1)$.

Such an α exists, because $\ell \mid \omega(\Theta^{\frac{\ell+1}{2}} f_\ell) = \frac{\ell^2-1}{2} + \frac{\ell+1}{2}(\ell + 1)$.

Definition 39. We define j to be the least natural number such that $\ell \mid \omega(\Theta^{\frac{\ell+3}{2}+j} f_\ell)$. Such a j exists, because $\ell \mid \omega(\Theta^{\ell-2} f_\ell)$.

Note : This definition requires that $(f_\ell|U) = 0$. We will assume this fact from now on.

Lemma 40. $\alpha \leq \frac{\ell+3}{2}$.

Proof.

□

Lemma 41. $j \leq \frac{\ell-7}{2}$.

Proof.

□

Lemma 42. For all $m \leq j$, $\omega(\Theta^{\frac{\ell+3}{2}+m} f_\ell) = \frac{\ell^2-1}{2} + (\frac{\ell+3}{2} + m)(\ell + 1) - (\alpha + 1)(\ell - 1)$.

Proof.

□

Lemma 43. $\ell \mid (j + 1) + (\alpha + 1)$.

Proof.

□

Lemma 44. $\alpha + 1 = \frac{\ell+5}{2}$.

Proof.

□

Theorem 45. $\omega(\Theta^{\frac{\ell+3}{2}} f_\ell) = \frac{\ell^2-1}{2} + 4$.

Proof.

□

Lemma 46. $f_\ell(\delta_\ell + 1) = 1$.

Proof.

□

Definition 47. Let p be the partion function. We say that there is a Ramanujan congruence mod a prime ℓ if $\forall n \in \mathbb{N}$, $\ell \mid p(\ell n - \delta_\ell)$.

Theorem 48. If $n > 0$ and $m \geq n$, then $p(n)$ is equal to the n th coefficient in the product exapansion of

$$\prod_{i=0}^m (1 - X^{i+1})^{-1}$$

Note : In lean, $p(0) = 0$.

Proof. This proof is almost entirely due to Archive.Wiedijk100Theorems.Partition

□

Definition 49. Let α be a commutative semiring, and let $f, g : \mathbb{N} \rightarrow \alpha[[X]]$ be two sequences of power series with coefficients in α . We say that f and g are eventually equal, and write $f \longrightarrow g$, if $\forall n \in \mathbb{N}$, $\exists m \in \mathbb{N}$, $\forall k \leq n$, $\forall j \geq m$, the k th coefficient of $f(j)$ is equal to the k th coefficient of $g(j)$. This is an equivalence relation.

Theorem 50. If $f \longrightarrow f'$ and $g \longrightarrow g'$, then $fg \longrightarrow f'g'$.

Proof.

□

Lemma 51. Let $n, N, \ell \in \mathbb{N}$ with $n < N$ and $\ell > 0$. Let $a : \mathbb{N} \rightarrow \alpha$ be a sequence with coefficients in a semiring α . Then the n th coefficient of

$$\sum_{i=0}^N a(n)X^{\ell i} = \begin{cases} a(n/\ell) & \text{if } \ell \mid n, \\ 0 & \text{otherwise.} \end{cases}$$

Proof. □

Lemma 52. Let $j, \ell, N, M \in \mathbb{N}$ with $\ell > 0$ and $N, M > \ell j$. Let $a, b : \mathbb{N} \rightarrow \alpha$. Then the ℓj th coefficient of

$$\sum_{i=0}^N a(i)X^i \sum_{i=0}^M b(i)X^{\ell i} = \sum_{x+y=j} a(\ell x)b(y)$$

Proof. □

Lemma 53. Let $m, N \in \mathbb{N}$. Let $f : \mathbb{N} \rightarrow \alpha[[X]]$ be a sequence with $f(0) = 0$, the zero power series. Then

$$X^m \sum_{i=0}^N f(i)X^i = \sum_{i=0}^{N+m} f(i-m)X^i$$

Proof. □

Lemma 54. Let $M, \ell, k \in \mathbb{N}$ with $\ell \nmid k$. Then the k th coefficient of

$$\prod_{i=0}^M (1 - X^{\ell(i+1)})^\ell = 0$$

Proof. □

Lemma 55. Let $\ell, K \in \mathbb{N}$ with $\ell > 0$. Then there exists a $c : \mathbb{N} \rightarrow \alpha$ and $M \in \mathbb{N}$ such that

$$\prod_{i=0}^K (1 - X^{\ell(i+1)})^\ell = \sum_{i=0}^M c(i)X^{\ell i}$$

Proof. □

Theorem 56. The coefficients of the Delta product are eventually equal to the coefficients of the Delta function. In other words,

$$X \prod_{i \leq \cdot} (1 - X^{i+1})^{24} \longrightarrow \sum_{i \leq \cdot} \Delta(i)X^i$$

Proof. □

Theorem 57. If $\ell \geq 5$ is prime, then the coefficients of the f_ℓ product are eventually equal to the coefficients of the f_ℓ function. In other words,

$$(X^{\delta_\ell} \prod_{i \leq \cdot} (1 - X^{i+1})^{24\delta_\ell} \longrightarrow \sum_{i \leq \cdot} f_\ell(i)X^i$$

Proof. □

Theorem 58. Let $\ell \geq 5$ be prime. If there is a ramanujan congruence mod ℓ , then $(f_\ell|U) = 0$.

Proof. □

Theorem 59. $E_4(1) = 240$.

Proof. □

Theorem 60. $G_{6\delta_\ell+2, \delta_\ell}(\delta_\ell + 1) \equiv 241 \pmod{\ell}$.

Proof. □

Theorem 61.

$$\Theta^{\frac{\ell+3}{2}} f_\ell \equiv \delta_\ell^{\frac{\ell+3}{2}} G_{6\delta_\ell+2, \delta_\ell} \pmod{\ell}.$$

Proof. □

Theorem 62. $(\Theta^{\frac{\ell+3}{2}} f_\ell)(\delta_\ell + 1) = 241(\delta_\ell + 1)^{\frac{\ell+3}{2}}.$

Proof. □

Theorem 63. *If $\ell \geq 13$ is prime, then $(f_\ell|U) \neq 0$.*

Proof. □

Theorem 64. *If $\ell \geq 13$ is prime, then there does not exist a ramanujan congruence mod ℓ .*

Proof. □